

Critical-path formulas single-cycle datapath, AY25/26 Q28 style. **Bold** = usually wins the max; grey = usually loses. Check with the given numbers.

var	IM	RF	ALU	DM	MUX	CTRL	ALUC	SE	SHL	AND/OR	ADD
name	Instruction Memory	Register File	Arithmetic Logic Unit	Data Memory	Multiplexer	Control unit	ALU Control	Sign Extend	Shift Left 2	logic gate	Adder (not the ALU)
does	reads the instruction at PC	reads rs, rt (RD1, RD2); writes rd/rt at the end	add / sub / and / or / slt; gives is0	lw reads, sw writes at the ALU address	picks one input by a select signal (any mux)	opcode → RegDst, ALUSrc, MemToReg, ALUop, ...	ALUop + funct → 4-bit ALU operation	16-bit imm → 32 bits (copies bit 15)	×4: word offset → byte offset	AND: PCSrc = Branch · is0	PC + 4, and PC+4 + offset×4
Tut 5 given	400	200	120	350	30	100	100	20	20	20	100

instr	full formula	expected	mod 1	mod 2	mod 3	Tut 5	total
R-type	IM + max(RF + MUX , CTRL + ALUC) + ALU + MUX + RF	IM + RF + MUX + ALU + MUX + RF				980	
lw	IM + max(RF , CTRL + ALUC , max(SE, CTRL) + MUX) + ALU + DM + MUX + RF	IM + RF + ALU + DM + MUX + RF				1300	
sw	IM + max(RF , CTRL + ALUC , max(SE, CTRL) + MUX) + ALU + DM	IM + RF + ALU + DM				1070	
beq	IM + max(RF + MUX , CTRL + ALUC) + ALU + AND, SE + SHL + ADD) + MUX	IM + RF + MUX + ALU + AND + MUX				800	
addi	IM + max(RF , CTRL + ALUC , max(SE, CTRL) + MUX) + ALU + MUX + RF	IM + RF + ALU + MUX + RF				950	
Extensions: each needs the mods listed. New ZE / gate / shift cost like SE; an extra mux = MUX. Adding the j hardware also puts one more MUX at the end of every other instruction's PC path (harmless unless beq/bne).							
bne	IM + max(RF + MUX , CTRL + ALUC) + ALU + AND + OR + MUX	IM + RF + MUX + ALU + AND + OR + MUX	signal BNE	NOT on is0, AND, OR into PCSrc		820	
sll	IM + max(RF + MUX , CTRL + ALUC) + ALU + MUX + RF	IM + RF + MUX + ALU + MUX + RF	shamt [10:6] into ALU	shifter in ALU + ALUctrl code		980	
srl	IM + max(RF + MUX , CTRL + ALUC) + ALU + MUX + RF	IM + RF + MUX + ALU + MUX + RF	shamt [10:6] into ALU	shifter in ALU + ALUctrl code		980	
nor	IM + max(RF + MUX , CTRL + ALUC) + ALU + MUX + RF	IM + RF + MUX + ALU + MUX + RF	ALUctrl row: funct 27 → 1100			980	
andi	IM + max(RF , CTRL + ALUC , max(ZE, CTRL) + MUX + MUX) + ALU + MUX + RF	IM + RF + ALU + MUX + RF	ZeroExtend unit	ExtOp mux before ALUSrc mux	ALUop code for AND	950	
ori	IM + max(RF , CTRL + ALUC , max(ZE, CTRL) + MUX + MUX) + ALU + MUX + RF	IM + RF + ALU + MUX + RF	ZeroExtend unit	ExtOp mux before ALUSrc mux	ALUop code for OR	950	
slti	IM + max(RF , CTRL + ALUC , max(SE, CTRL) + MUX) + ALU + MUX + RF	IM + RF + ALU + MUX + RF	ALUop code for slt (e.g. 11)			950	
lui	IM + max(SHL, CTRL) + MUX + RF	IM + CTRL + MUX + RF	imm ≪ 16 unit	3rd MemToReg mux input		730	
j	IM + max(SHL, CTRL) + MUX	IM + CTRL + MUX	ShiftL2 on [25:0] + PC+4[31:28]	Jump mux after PCSrc mux	signal Jump	530	
jr	IM + max(RF , CTRL) + MUX	IM + RF + MUX	RD1 into a new PC mux	signal JR		630	
jal	IM + max(CTRL + MUX + RF , max(SHL, CTRL) + MUX)	IM + CTRL + MUX + RF	the j hardware	RegDst mux input 31	MemToReg mux input PC+4	730	
clock period = slowest instruction						1300	

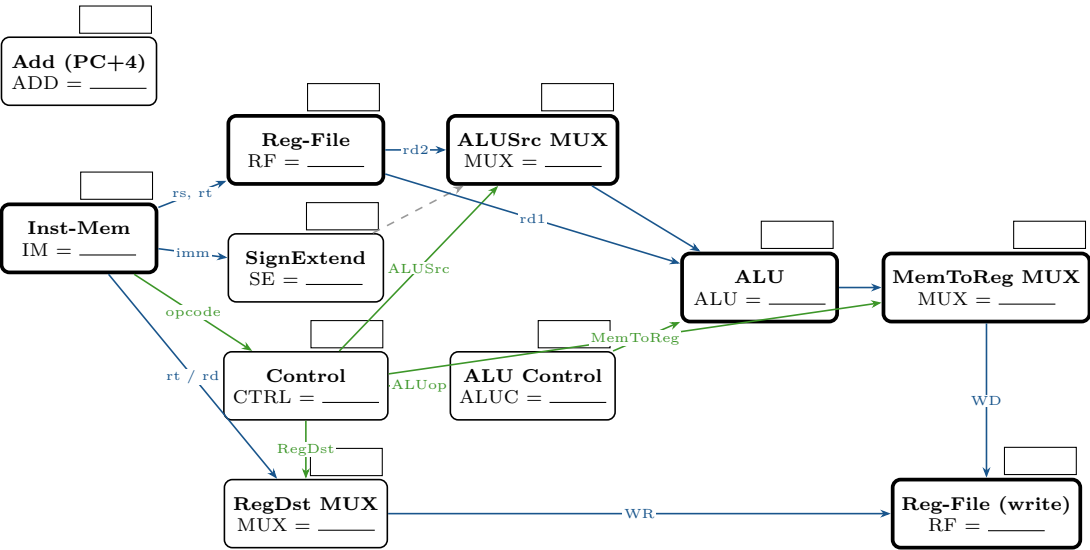
Rules. finish = max(inputs) + own latency. R / beq: the ALUSrc mux takes RD2, so it sits *after* RF. lw / sw / addi: the mux takes SE (early), so it's RF vs CTRL + ALUC. A mux also waits for its select from Control (why j = max(SHL, CTRL) + MUX). **Adder never counts:** PC+4 starts at $t = 0$ beside IM.

Tick questions (Q28): a column is on the critical path iff its variable appears in the *expected* formula; the number of times it appears = how many times (addi: MUX once, RF twice). If two terms tie in a max (Tut 5: RF = CTRL + ALUC = 200), both paths are critical.

Critical-path graphs AY25/26 Q28 style: plug in the latencies the question gives

Variables: IM Inst-Mem, RF Reg-File (read and write), MUX any mux, ALU, DM Data-Mem, CTRL Control, ALUC ALU Control, SE SignExtend, ZE ZeroExtend, SHL shift, AND / OR gates, ADD any adder (not the ALU). **blue** = data, **green** = control, dashed = mux input not selected (does not count). Tut 5 (ps): IM 400, RF 200, ALU 120, DM 350, MUX 30, CTRL 100, ALUC 100, SE / SHL / AND 20, ADD 100.

A. R-type add \$rd, \$rs, \$rt



Total = _____ ps

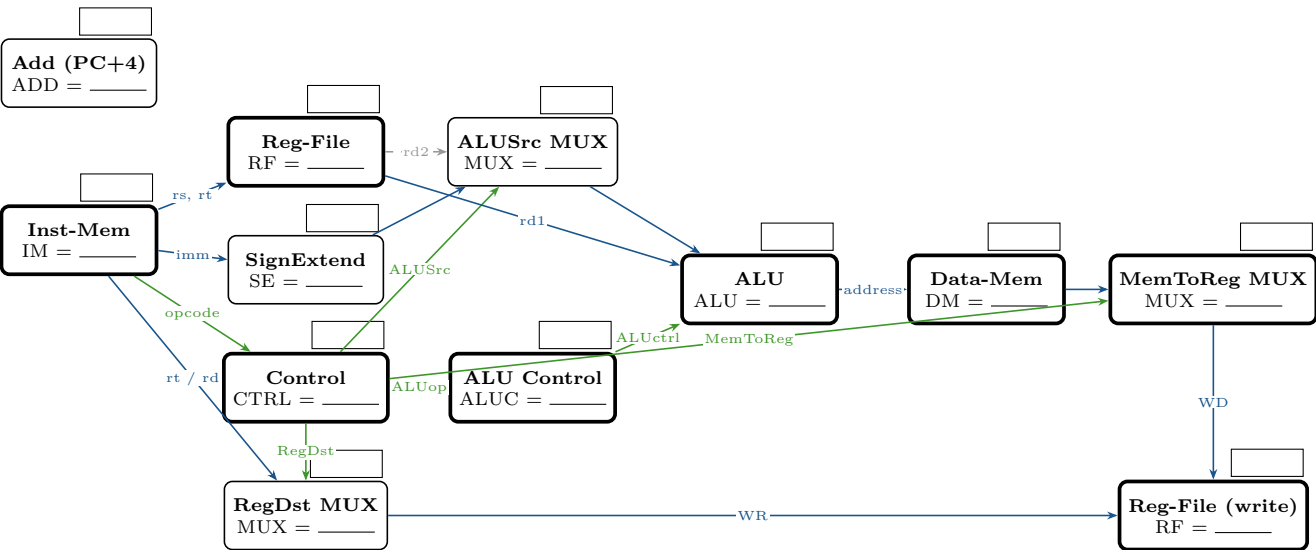
Formula (**bold** = usually wins the max):

$$IM + \max(\mathbf{RF} + \mathbf{MUX}, \mathbf{CTRL} + \mathbf{ALUC}) + \mathbf{ALU} + \mathbf{MUX} + \mathbf{RF}$$

How: write each given latency in its box; a box finishes at max(finish of its *solid* inputs) + its own latency; dashed inputs do not count. Thick border = usual critical path, but check every max with the real numbers.

Tut 5 check: Add (PC+4) 100, Inst-Mem 400, SignExtend 420, Control 500, RegDst MUX 530, Reg-File 600, ALU Control 600, ALUSrc MUX 630, ALU 750, MemToReg MUX 780, Reg-File (write) 980. **Total 980 ps.**

B. lw lw \$rt, imm(\$rs)



Total = _____ ps

Formula (**bold** = usually wins the max):

$$IM + \max(\mathbf{RF}, \mathbf{CTRL} + \mathbf{ALUC}, \max(\mathbf{SE}, \mathbf{CTRL}) + \mathbf{MUX}) + \mathbf{ALU} + \mathbf{DM} + \mathbf{MUX} + \mathbf{RF}$$

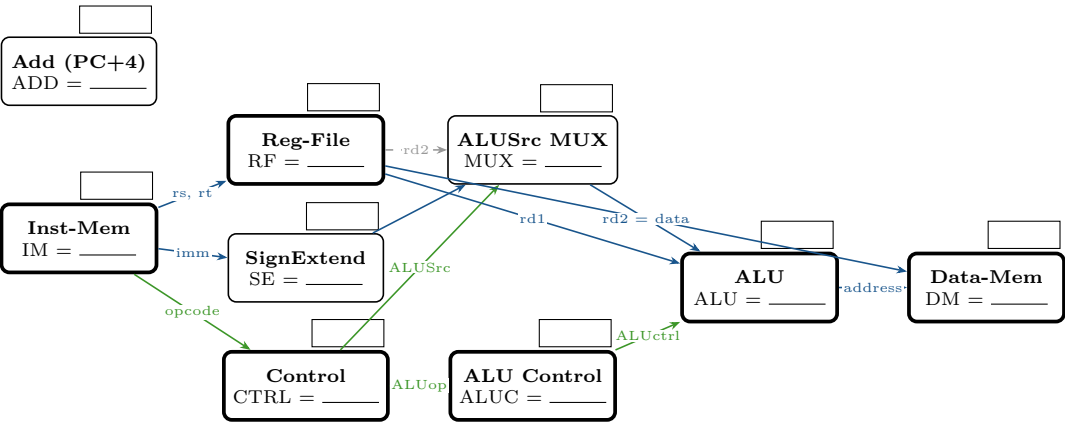
How: write each given latency in its box; a box finishes at max(finish of its *solid* inputs) + its own latency; dashed inputs do not count. Thick border = usual critical path, but check every max with the real numbers.

Tut 5 check: Add (PC+4) 100, Inst-Mem 400, SignExtend 420, Control 500, RegDst MUX 530, ALUSrc MUX 530, Reg-File 600, ALU 720, Data-Mem 1070, MemToReg MUX 1100, Reg-File (write) 1300. **Total 1300 ps.**

Critical-path graphs AY25/26 Q28 style: plug in the latencies the question gives

Variables: IM Inst-Mem, RF Reg-File (read and write), MUX any mux, ALU, DM Data-Mem, CTRL Control, ALUC ALU Control, SE SignExtend, ZE ZeroExtend, SHL shift, AND / OR gates, ADD any adder (not the ALU). **blue** = data, **green** = control, dashed = mux input not selected (does not count). Tut 5 (ps): IM 400, RF 200, ALU 120, DM 350, MUX 30, CTRL 100, ALUC 100, SE / SHL / AND 20, ADD 100.

C. **sw** sw \$rt, imm(\$rs)



Total = _____ ps

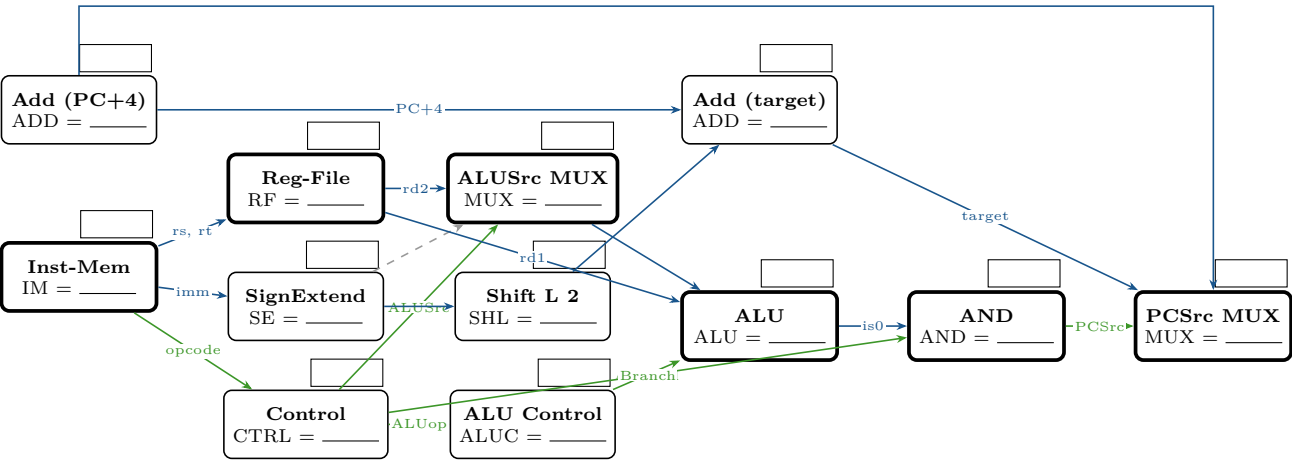
Formula (**bold** = usually wins the max):

$$IM + \max(\mathbf{RF}, \mathbf{CTRL} + \mathbf{ALUC}, \max(\mathbf{SE}, \mathbf{CTRL}) + \mathbf{MUX}) + \mathbf{ALU} + \mathbf{DM}$$

How: write each given latency in its box; a box finishes at max(finish of its *solid* inputs) + its own latency; dashed inputs do not count. Thick border = usual critical path, but check every max with the real numbers.

Tut 5 check: Add (PC+4) 100, Inst-Mem 400, SignExtend 420, Control 500, ALUSrc MUX 530, Reg-File 600, ALU Control 600, ALU 720, Data-Mem 1070. **Total 1070 ps.**

D. **beq** beq \$rs, \$rt, imm



Total = _____ ps

Formula (**bold** = usually wins the max):

$$IM + \max(\max(\mathbf{RF} + \mathbf{MUX}, \mathbf{CTRL} + \mathbf{ALUC}) + \mathbf{ALU} + \mathbf{AND}, \mathbf{SE} + \mathbf{SHL} + \mathbf{ADD}) + \mathbf{MUX}$$

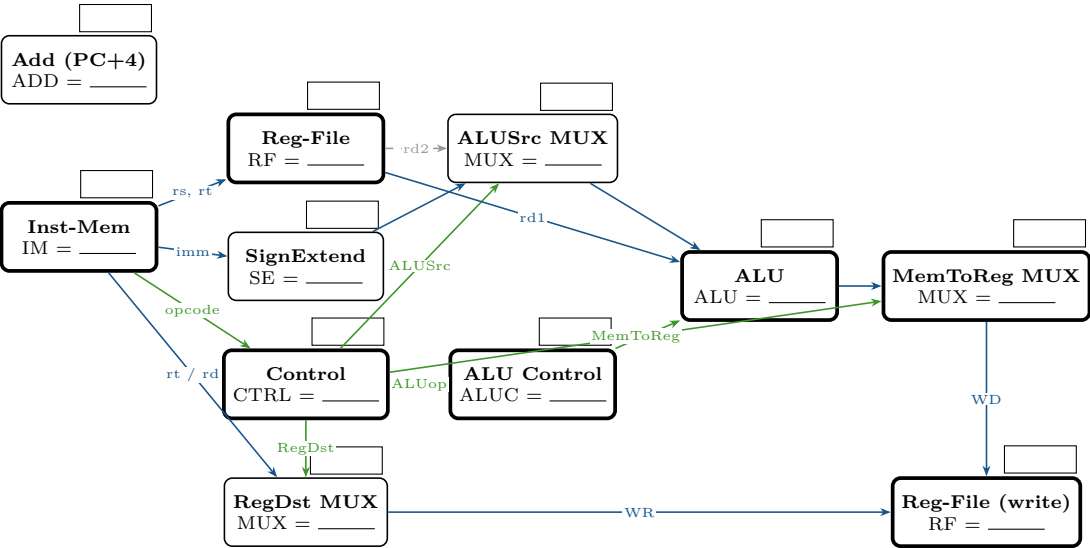
How: write each given latency in its box; a box finishes at max(finish of its *solid* inputs) + its own latency; dashed inputs do not count. Thick border = usual critical path, but check every max with the real numbers.

Tut 5 check: Add (PC+4) 100, Inst-Mem 400, SignExtend 420, Shift L 2 440, Control 500, Add (target) 540, Reg-File 600, ALU Control 600, ALUSrc MUX 630, ALU 750, AND 770, PCSrc MUX 800. **Total 800 ps.**

Critical-path graphs AY25/26 Q28 style: plug in the latencies the question gives

Variables: IM Inst-Mem, RF Reg-File (read and write), MUX any mux, ALU, DM Data-Mem, CTRL Control, ALUC ALU Control, SE SignExtend, ZE ZeroExtend, SHL shift, AND / OR gates, ADD any adder (not the ALU). **blue** = data, **green** = control, dashed = mux input not selected (does not count). Tut 5 (ps): IM 400, RF 200, ALU 120, DM 350, MUX 30, CTRL 100, ALUC 100, SE / SHL / AND 20, ADD 100.

E. addi addi \$rt, \$rs, imm



Total = _____ ps

Formula (**bold** = usually wins the max):

$$IM + \max(\mathbf{RF}, \mathbf{CTRL} + \mathbf{ALUC}, \max(\mathbf{SE}, \mathbf{CTRL}) + \mathbf{MUX}) + \mathbf{ALU} + \mathbf{MUX} + \mathbf{RF}$$

How: write each given latency in its box; a box finishes at max(finish of its *solid* inputs) + its own latency; dashed inputs do not count. Thick border = usual critical path, but check every max with the real numbers.

Tut 5 check: Add (PC+4) 100, Inst-Mem 400, SignExtend 420, Control 500, RegDst MUX 530, ALUSrc MUX 530, Reg-File 600, ALU Control 600, ALU 720, MemToReg MUX 750, Reg-File (write) 950. **Total 950 ps.**